

Fracture-aware Tessellation of Subdivision Surfaces

Brent Burley

Walt Disney Animation Studios
Burbank, USA

brent.burley@disneyanimation.com

Francisco Rodriguez

Walt Disney Animation Studios
Burbank, USA

francisco.rodriguez@disneyanimation.com

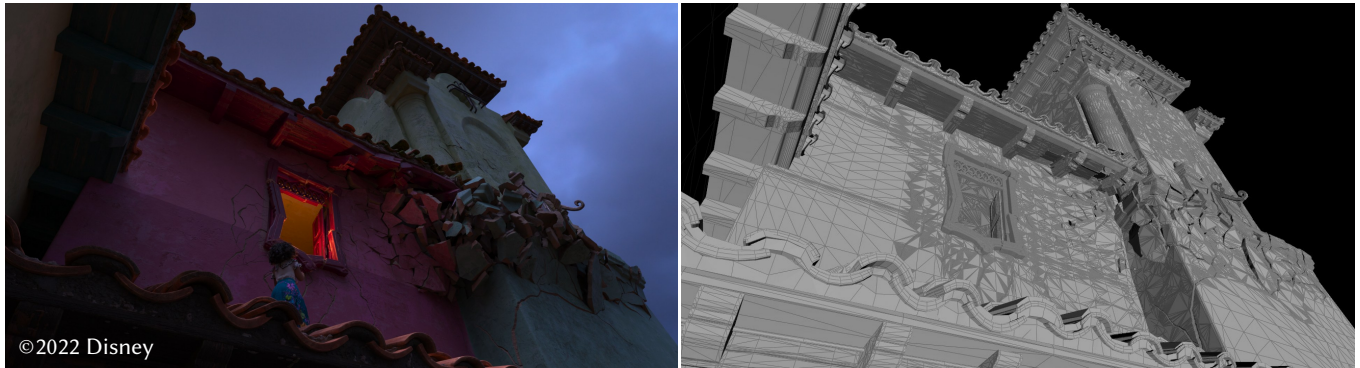


Figure 1: The destruction of Casa Madrigal in Disney Animation’s Encanto was rendered using fracture-aware tessellation; the fractured subdivision meshes provided to the renderer are shown on the right.

ABSTRACT

We introduce a new tessellation algorithm for production rendering of fractured subdivision surfaces that produces higher quality results with less distortion than previous approaches. Our tessellator is provided with a fractured subdivision control mesh along with the corresponding unfractured mesh. We use the unfractured mesh to evaluate limit positions of the fractured mesh vertices before tessellation, and we apply displacements at this time. During tessellation, we iteratively split edges, evaluating displaced limit positions at added points. Additionally, we measure deformations that were performed during the fracture process and apply these during tessellation to allow effects like animated crack spreading. Our approach produces seamless results with no distortion of the unfractured surface, and enables a more efficient fracture pipeline.

CCS CONCEPTS

• **Computing methodologies** → **Rendering**; *Mesh geometry models*.

ACM Reference Format:

Brent Burley and Francisco Rodriguez. 2022. Fracture-aware Tessellation of Subdivision Surfaces. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH ’22 Talks)*, August 07–11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532836.3536262>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH ’22 Talks, August 07–11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9371-3/22/08.

<https://doi.org/10.1145/3532836.3536262>



Figure 2: Crack lines are able to be revealed gradually while preserving the original detail of the surfaces. All of the cracks are in place at the start of the animation, and only the deformation is animated.

1 INTRODUCTION

The destruction of Casa Madrigal was a key story point in Disney Animation’s Encanto and included a large number of effects shots with everything from gradually spreading crack lines to the wholesale destruction of the house as seen in Figures 1 and 2. Given the fact that the house is a hero element comprised of subdivision surfaces with highly detailed displacements, producing these effects using existing methods would have been extremely challenging. In order to represent the detail of the unfractured house, surfaces would typically have to be pre-subdivided two or more levels, increasing the weight of the simulation meshes and reducing artist iteration times. Even with the added weight, the result would still only approximate the detail level achieved with ordinary render-time adaptive tessellation and displacement.

To address this challenge, we developed a new fracture-aware adaptive tessellation algorithm that allows fracturing a lightweight subdivision control mesh while retaining all the rendered detail of the original unfractured mesh.



Figure 3: A fractured boulder in Disney Animation’s *Encanto* rendered using our method. When the fragments are together (left), the seams are invisible and the render is indistinguishable from a render of the unfractured surface. When the fragments are separated (right), sharp edges and inside surface details including displacements can be observed.

2 ALGORITHM

Input. We start with a fractured subdivision control mesh and corresponding unfractured mesh. We separate the input mesh into connected fragments and tessellate each fragment individually.

Fragment preprocessing. For each fragment, we first compute a best-fit orthonormal transform [Horn et al. 1988] to map undeformed reference points (PRef) to their corresponding points in deformed fragment space; this transform accounts for any rigid animation that may have been applied to the fragment after fracturing. We then measure deformation vectors as the difference between the fragment point positions and PRef points transformed to fragment space, and we measure the effect of these deformations on the surface normals, expressed as a quaternion rotation at each point. We compute displaced limit positions of the fractured points by evaluating the unfractured control mesh [Stam 1998], applying the fragment transform, and evaluating the displacement shader.

Inside surface faces added during fracture have no correspondence to the unfractured subdivision mesh and thus have no predefined limit surface. To minimize unwanted stretching of the inside surface when the outside surface is projected to its limit position, the inside points are adjusted by the average movement of the surrounding points.

Inside edges shared by three or more fragments are tagged as sharp edges.

Tessellation. The fractured mesh is assumed to have been triangulated during the fracture process. We adaptively refine the tessellation by iteratively splitting edges of the mesh that are longer than the target length. We evaluate and displace added points as described above, and add interpolated deformation vectors and deformation rotations to the result. In order to produce a smooth tessellation for inside surfaces, Loop subdivision rules are applied to neighborhoods of added inside points; sharp inside edges are not smoothed.

Normals. Smooth normals are computed from the final tessellated mesh. To preserve sharp fracture edges, normals are not smoothed across inside/outside boundary edges, or sharp inside edges. To avoid a faceted appearance introduced by the linear interpolation of deformation vectors, we compute these normals from the undeformed surface and apply the interpolated deformation rotations to the normals.

3 RESULTS AND DISCUSSION

Our new fracture-aware tessellation method improved the render quality of our destruction elements by faithfully rendering all the detail in the original meshes, and at the same time allowed a lighter weight fracture simulation process with simpler setup and no requirement to pre-subdivide and apply displacements. Artists may still apply a level or two of uniform subdivision if the original mesh is too coarse for their simulation needs.

Before we implemented our new method, we had considered performing render-time subdivision of the fractured mesh (with subdivision crease tags applied to crack edges), but this would have added significant distortion to the mesh due to the topology change of the subdivision mesh, especially given the large number of added extraordinary points. In contrast, our method introduces no distortion because we evaluate the limit positions using the unfractured mesh. Doing so also makes our method robust to degenerate (non-manifold) fracture meshes which frequently occur during fracturing and otherwise would be unable to be subdivided.

REFERENCES

- Berthold K. P. Horn, Hugh M. Hilden, and Shahriar Negahdaripour. 1988. Closed-form solution of absolute orientation using orthonormal matrices. *J. Opt. Soc. Am. A* 5, 7 (Jul 1988), 1127–1135. <https://doi.org/10.1364/JOSAA.5.001127>
- Jos Stam. 1998. Exact Evaluation of Catmull-Clark Subdivision Surfaces at Arbitrary Parameter Values. In *Proceedings of the 25th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '98)*. Association for Computing Machinery, New York, NY, USA, 395?404. <https://doi.org/10.1145/280814.280945>